

KUNAL SINGH

Patna, Bihar, India | +91-7870638125 | singhkunal1642@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [LeetCode](#) | [GeeksforGeeks](#)

Education

National Institute of Technology, Raipur

Master of Computer Applications (MCA) – CGPA: 7.51

Raipur, Chhattisgarh, India

Aug 2024 – Present

Indira Gandhi National Open University

Bachelor of Computer Applications (BCA) – Percentage: 63.54%

New Delhi, India

Jan 2021 – Jan 2024

Technical Skills

Languages: JavaScript (ES6+), Python, C++

Backend Development: Node.js, Express.js, REST APIs

Frontend Development: React.js, Tailwind CSS, Bootstrap CSS, HTML5, CSS3

AI & LLM Integration: Groq API, Gemini API, Prompt Engineering

Databases & ORM: MongoDB (Mongoose), SQL

DevOps & Tools: Docker, Docker Compose, Nginx, Git, GitHub, Postman

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Open Source Experience

GirlScript Summer of Code (GSSoC)

Remote

Open Source Contributor

Jul 2025 – Sep 2025

- Identified, reproduced, and resolved bugs across open-source web repositories, collaborating directly with maintainers.
- Submitted modular frontend and backend pull requests adhering to structured Git/GitHub code review workflows.
- Contributed to issue triaging, code refactoring, and documentation improvements across distributed codebases.

Projects

Intervue — Real-Time Technical Interview Platform | *React.js, Node.js, Express, Stream SDK, Monaco, Inngest, MongoDB* 2026

- Architected a real-time mock interview platform featuring **low-latency video calling, in-session chat, and collaborative code editing** powered by Stream Video SDK, Stream Chat, and Monaco Editor.
- Integrated the **JDoodle API** for multi-language sandboxed code execution (JavaScript, Python, Java), allowing candidates and interviewers to run and evaluate DSA solutions during live sessions.
- Implemented **Clerk authentication** paired with custom Express middleware to auto-synchronize user profiles into MongoDB and Stream on initial login.
- Designed an **event-driven user lifecycle** using Inngest serverless workflows with automated retry queues to reliably process Clerk webhook events.
- Built session lifecycle management with **host-only controls**, role-based access, and automatic teardown of Stream channels on interview termination. [[GitHub](#) | [Demo](#)]

Expense Tracker — AI-Enhanced Financial Management | *MERN Stack, Groq API, Gemini API, Recharts, MongoDB* 2025 – 2026

- Built a full-stack financial platform with **JWT access tokens + HTTP-only cookie refresh token rotation** (SHA-256 hashed sessions in MongoDB), OTP email verification via Brevo, and a role-based admin dashboard.
- Implemented **keyword-based transaction categorization** across 22+ categories and an **AI financial assistant** powered by Groq API with Gemini fallback for natural language spending queries.
- Engineered **configurable budget threshold alerts** (daily/weekly/monthly/yearly) using **node-cron** with period-aware deduplication to prevent duplicate alert notifications.
- Developed an interactive analytics dashboard using **Recharts** (Area, Bar, Pie charts) with client-side **PDF and CSV export** functionality via jsPDF. [[GitHub](#) | [Demo](#)]

Note Vault — Containerized Secure Note Service | *React.js, Node.js, Express, MongoDB, Docker, Nginx, Jest, Supertest* 2026

- Implemented an authentication service featuring **HTTP-only cookie refresh token rotation** (SHA-256 hashed active sessions in MongoDB) and instant session revocation on logout.
- Developed **RESTful APIs** supporting note CRUD operations, multi-field regex search (tags and title), soft-delete with timestamped trash bin, and instant note restoration.
- Fortified endpoints with **per-user and per-IP rate limiting (30 req/min)** using **express-rate-limit** and input validation middleware.
- Containerized the frontend and backend using **Docker and Docker Compose**, configuring a multi-stage build with **Nginx** to serve static React assets and route SPA traffic.
- Constructed **integration test suites** with **Jest and Supertest** backed by an in-memory MongoDB server, validating auth lifecycles and CRUD edge cases. [[GitHub](#) | [Demo](#)]

Certifications & Achievements

- Postman API Fundamentals Student Expert
- GirlScript Summer of Code 2025 Contributor
- GeeksforGeeks Python Course (2025)

- HackerRank Problem Solving (Basic)
- 160 Days DSA Challenge
- ET AutoTech Hackathon 2026